

3. DATA EXPLORATION (2)

SECTION 3.5~3.7



CONTENTS

- **Advanced Data Exploration**
 - Visualizing Relationships Between Features
 - Measuring Covariance & Correlation
- **Data Preparation**
 - Normalization
 - Binning
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- **Summary**

Table 3.7 The details of a professional basketball team.

ID	POSITION	HEIGHT	WEIGHT	CAREER STAGE	AGE	SPONSORSHIP EARNINGS	SHOE SPONSOR
1	forward	192	218	veteran	29	561	yes
2	center	218	251	mid-career	35	60	no
3	forward	197	221	rookie	22	1,312	no
4	forward	192	219	rookie	22	1,359	no
5	forward	198	223	veteran	29	362	yes
6	guard	166	188	rookie	21	1,536	yes
7	forward	195	221	veteran	25	694	no
8	guard	182	199	rookie	21	1,678	yes
9	guard	189	199	mid-career	27	385	yes
10	forward	205	232	rookie	24	1,416	no
11	center	206	246	mid-career	29	314	no
12	guard	185	207	rookie	23	1,497	yes
13	guard	172	183	rookie	24	1,383	yes
14	guard	169	183	rookie	24	1,034	yes
15	guard	185	197	mid-career	29	178	yes
16	forward	215	232	mid-career	30	434	no
17	guard	158	184	veteran	29	162	yes
18	guard	190	207	mid-career	27	648	yes
19	center	195	235	mid-career	28	481	no
20	guard	192	200	mid-career	32	427	yes
21	forward	202	220	mid-career	31	542	no
22	forward	184	213	mid-career	32	12	no
23	forward	190	215	rookie	22	1,179	no
24	guard	178	193	rookie	21	1,078	no
25	guard	185	200	mid-career	31	213	yes
26	forward	191	218	rookie	19	1,855	no
27	center	196	235	veteran	32	47	no
28	forward	198	221	rookie	22	1,409	no
29	center	207	247	veteran	27	1,065	no
30	center	201	244	mid-career	25	1,111	yes

VISUALIZING RELATIONSHIPS BETWEEN FEATURES

- A **scatter plot** is based on two axes: the horizontal axis represents one feature and the vertical axis represents a second.
- Each instance in a dataset is represented by **a point** on the plot determined by the values for that instance of the two features involved.

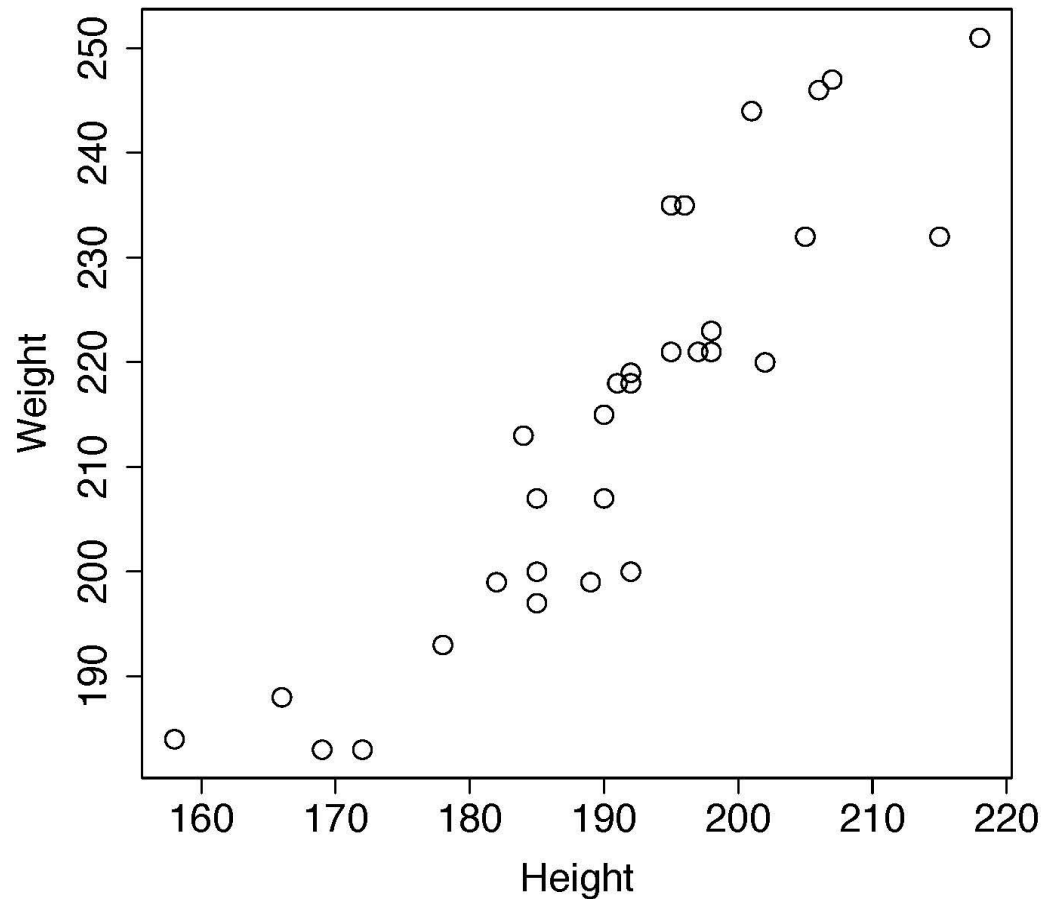


Figure 3.5-1: An example scatter plot showing the **strong positive covariance** between the **HEIGHT** and **WEIGHT** features from the professional basketball squad dataset in Table 3.7.

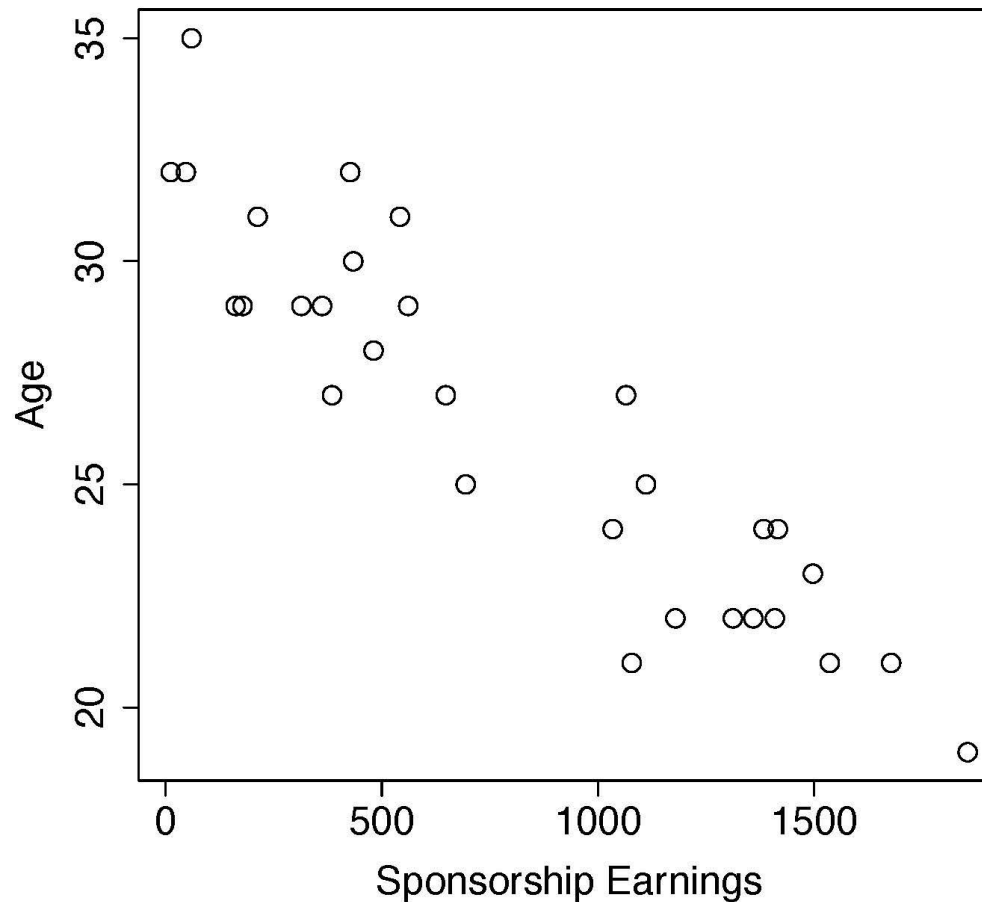


Figure 3.5-2: an example scatter plot showing the **strong negative covariance** between the **SPONSORSHIP EARNINGS** and **AGE** features from the dataset in Table 3.7

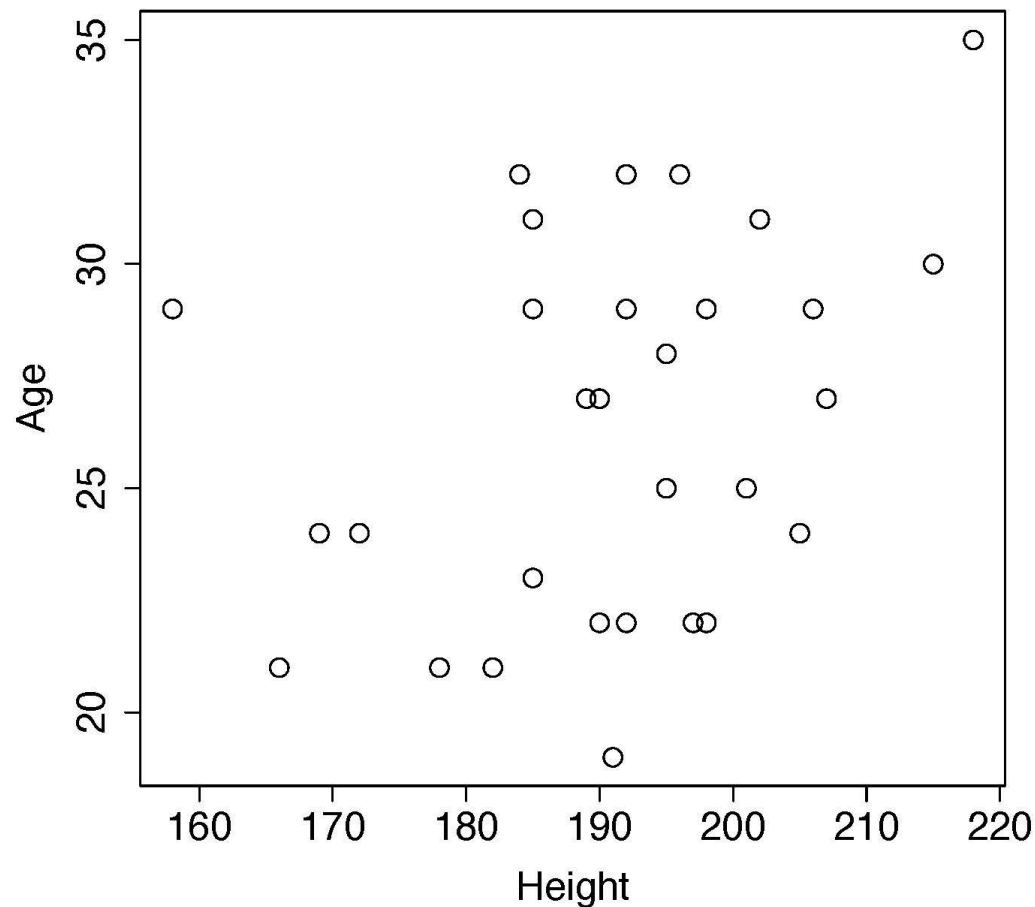


Figure 3.5-3: An example scatter plot showing the **lack of strong covariance** between the **HEIGHT** and **AGE** features from the dataset in Table 3.7

VISUALIZING RELATIONSHIPS BETWEEN FEATURES

- A **scatter plot matrix (SPLOM)** shows scatter plots for a whole collection of features arranged into a matrix.
- This is useful for exploring the relationships between groups of features - for example all of the **continuous** features in an ABT.

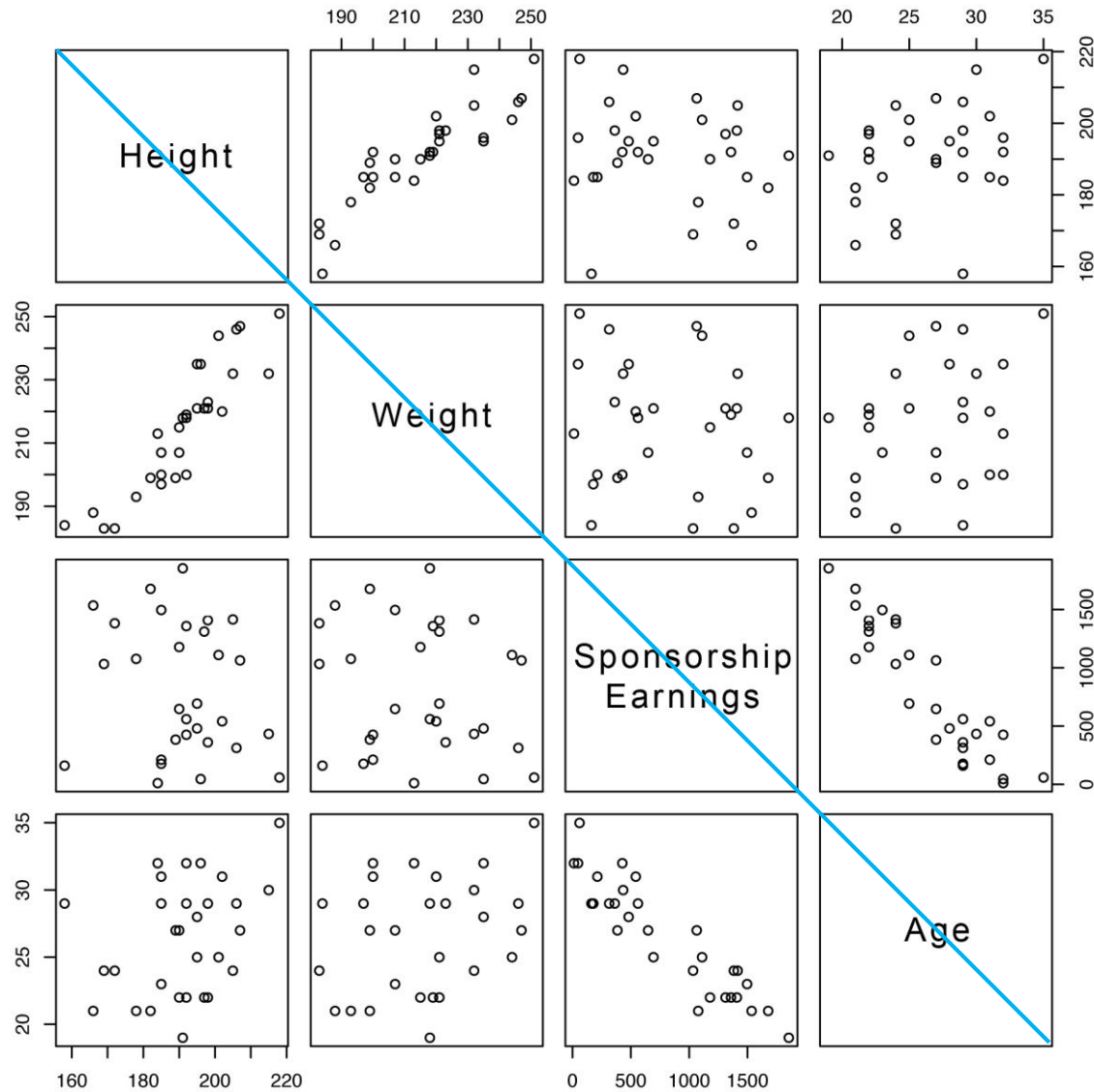


Figure3.6: A scatter plot matrix showing scatter plots of the continuous features from the professional basketball squad dataset.

VISUALIZING RELATIONSHIPS BETWEEN FEATURES

- The simplest way to visualize the relationship between two **categorical** variables is to use a collection of **small multiple bar plots**.

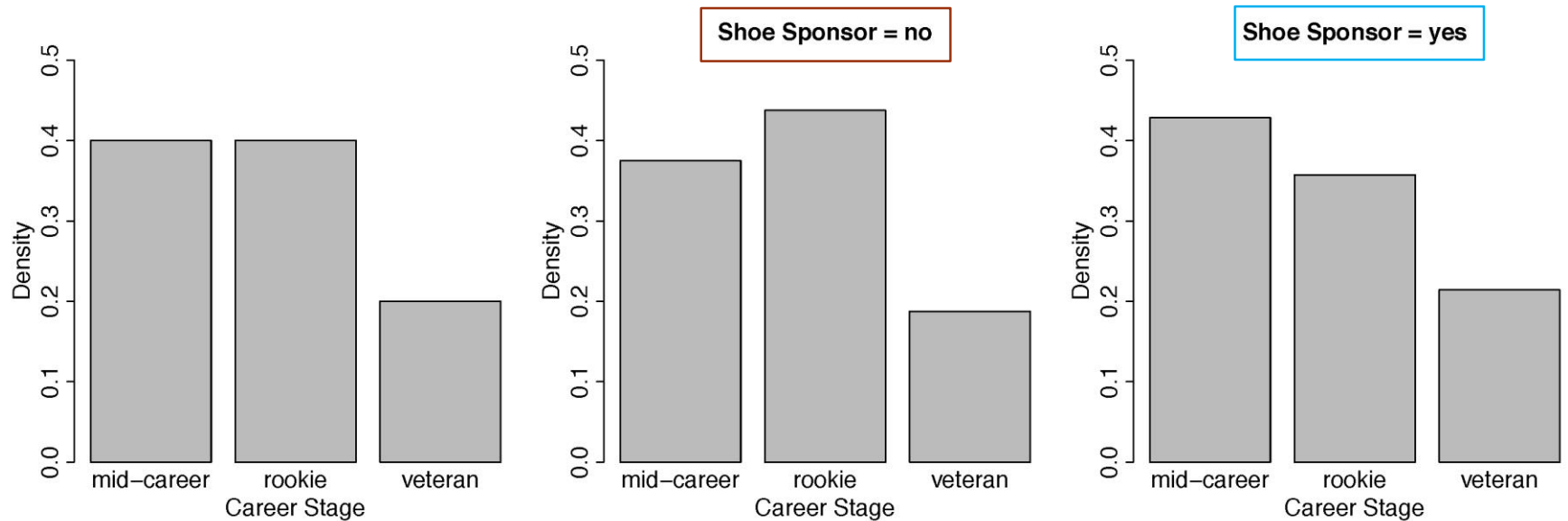


Figure 3.7-1: Using small multiple bar plot visualizations to illustrate the relationship between the **CAREER STAGE** and **SHOE SPONSOR** features.

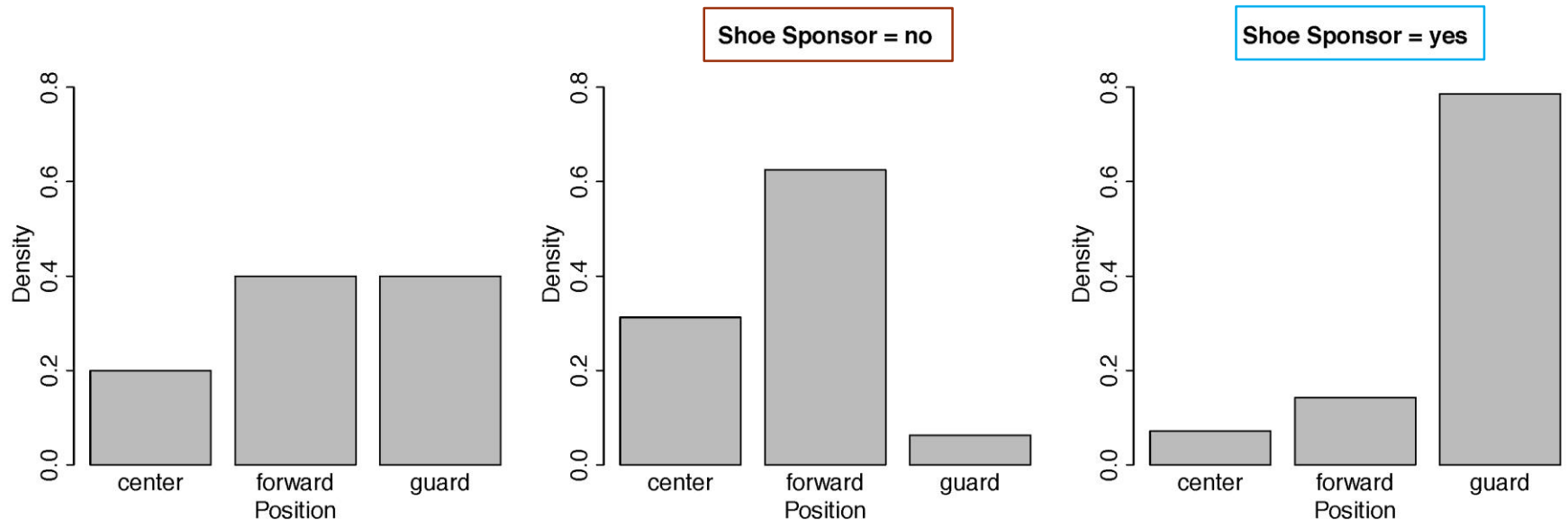
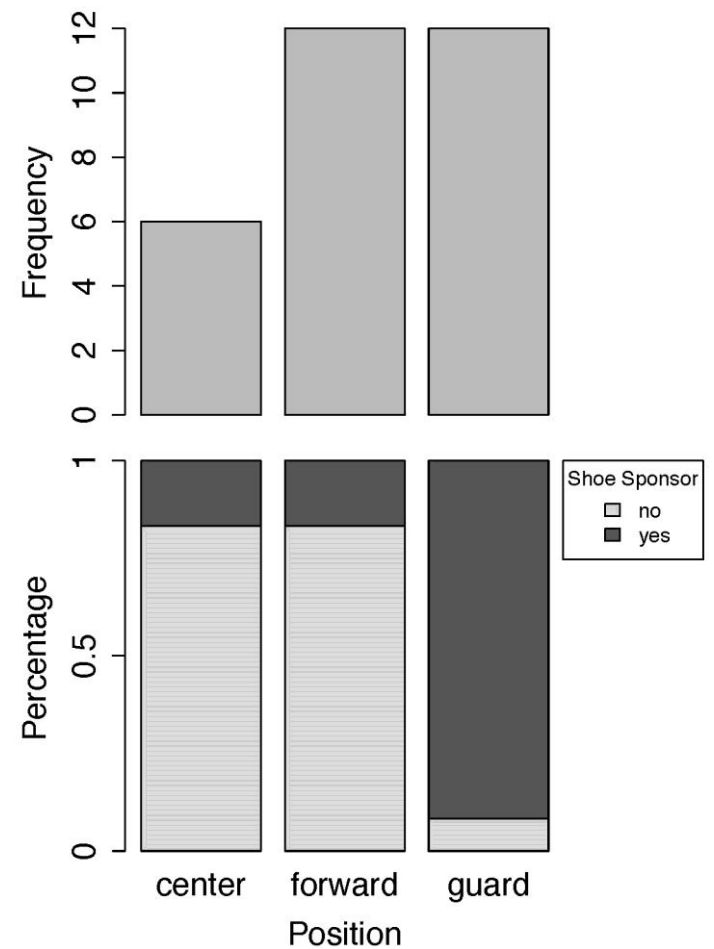
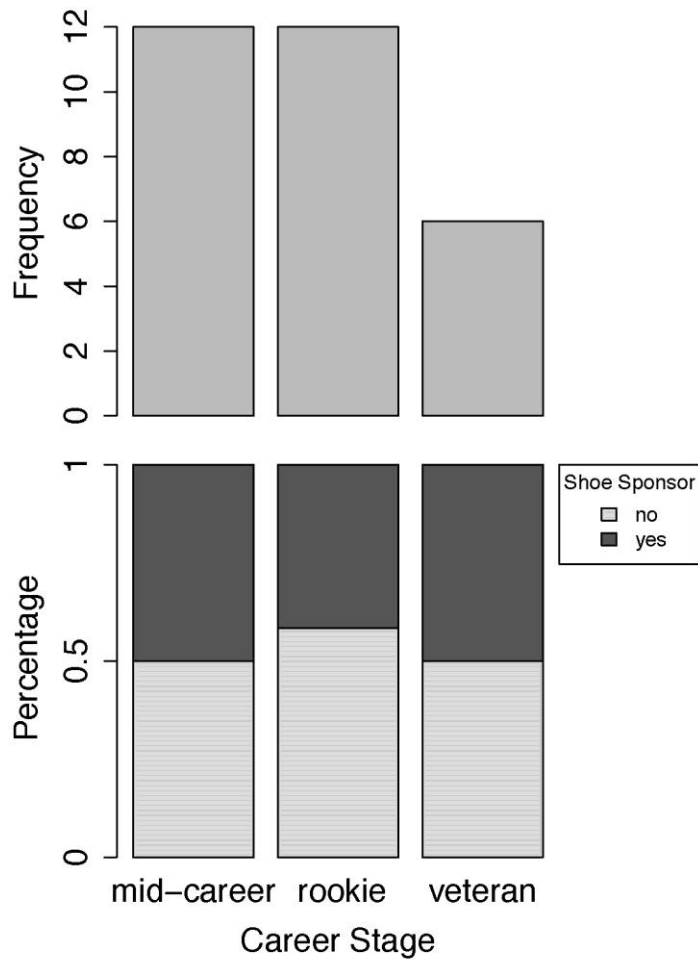


Figure 3.7-2: Using small multiple bar plot visualizations to illustrate the relationship between the **POSITION** and **SHOE SPONSOR** features.

VISUALIZING RELATIONSHIPS BETWEEN FEATURES

- If the number of levels of one of the features being compared is **no more than three**, we can use **stacked bar plots** as an **alternative** to the small multiples approach.



(a) Career Stage and Shoe Sponsor

(b) Position and Shoe Sponsor

Figure 3.8: Stacked bar plot visualizations.

VISUALIZING RELATIONSHIPS BETWEEN FEATURES

- To visualize the relationship between a **continuous** feature and a **categorical** feature, a **small multiples** approach that draws a **histogram** of the values of the continuous feature for each level of the categorical feature is useful.

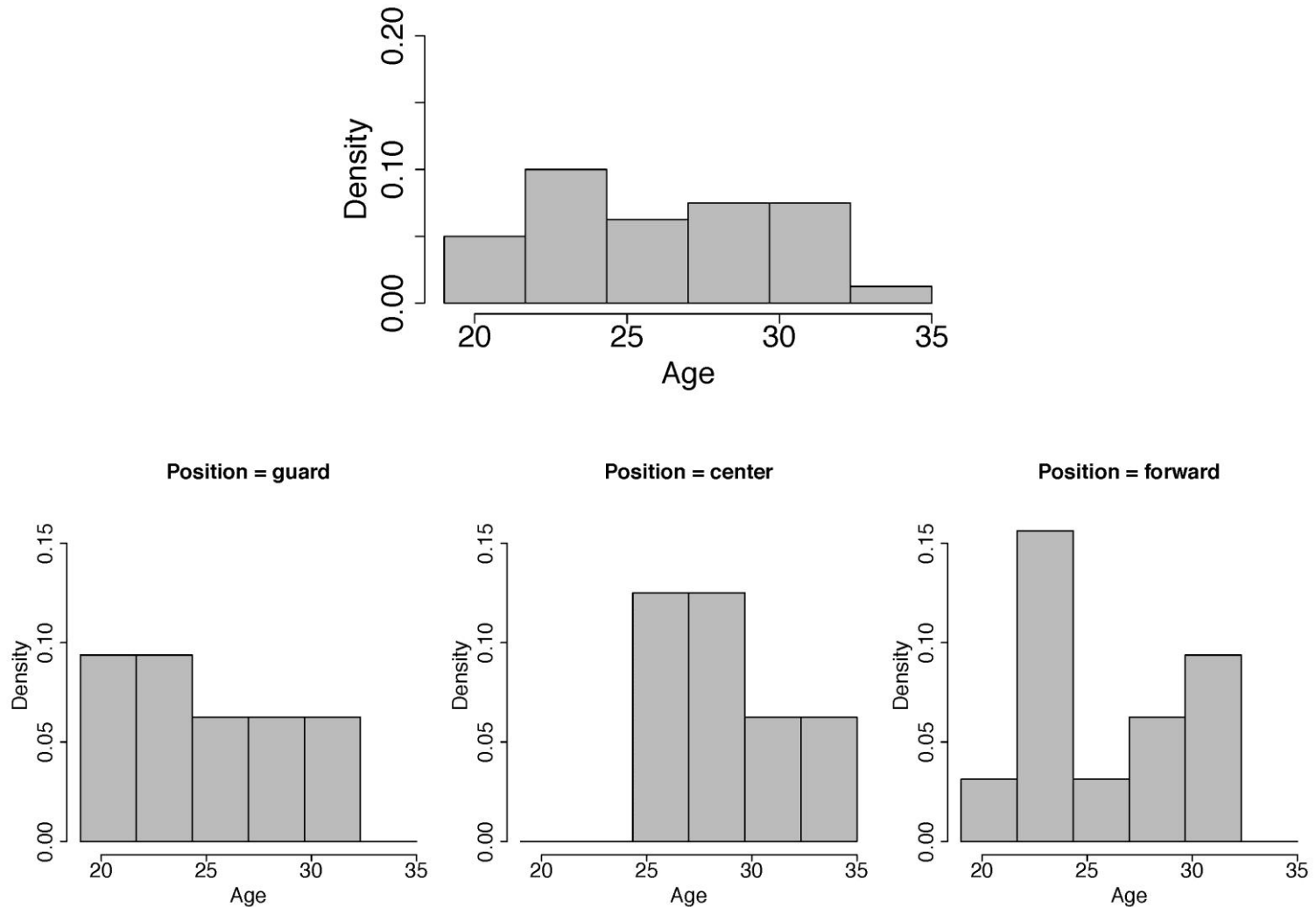
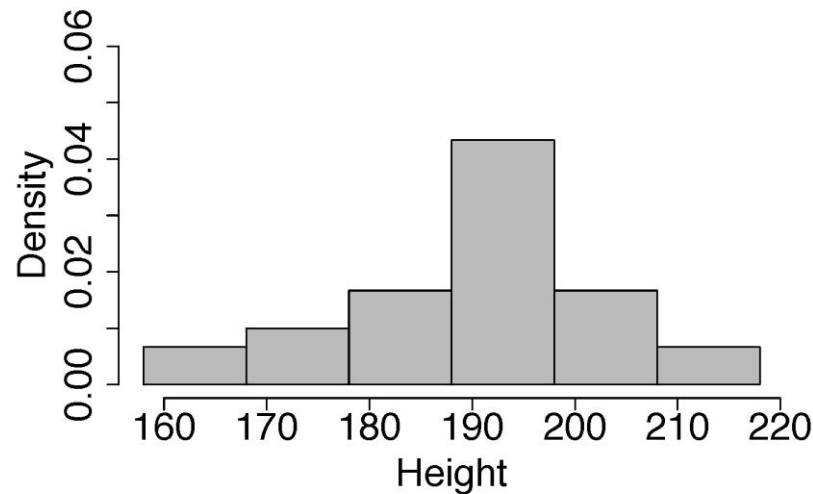
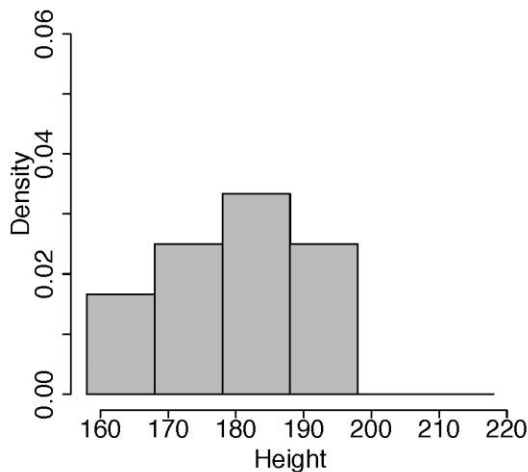


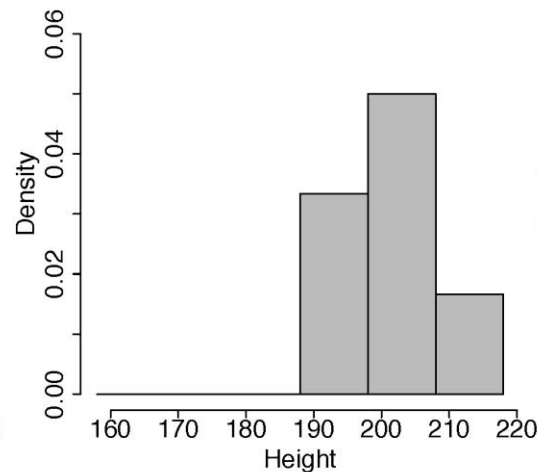
Figure 3.9-1: Using small multiple histograms to visualize the relationship between the **AGE** feature and the **POSITION** FEATURE.



Position = guard



Position = center



Position = forward

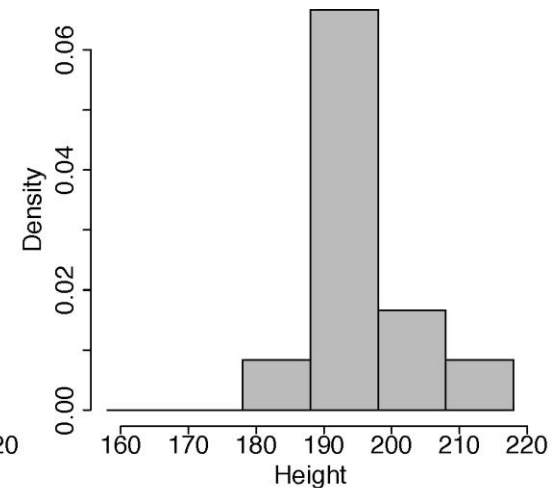


Figure 3.9-2: Using small multiple histograms to visualize the relationship between the **HEIGHT** feature and the **POSITION** feature.

VISUALIZING RELATIONSHIPS BETWEEN FEATURES

- A second approach to visualizing the relationship between a **categorical** feature and a **continuous** feature is to use a collection of **box plots**.
- For each level of the categorical feature a box plot of the corresponding values of the continuous feature is drawn.

FEATURE VALUES

Values displayed
for a single
feature

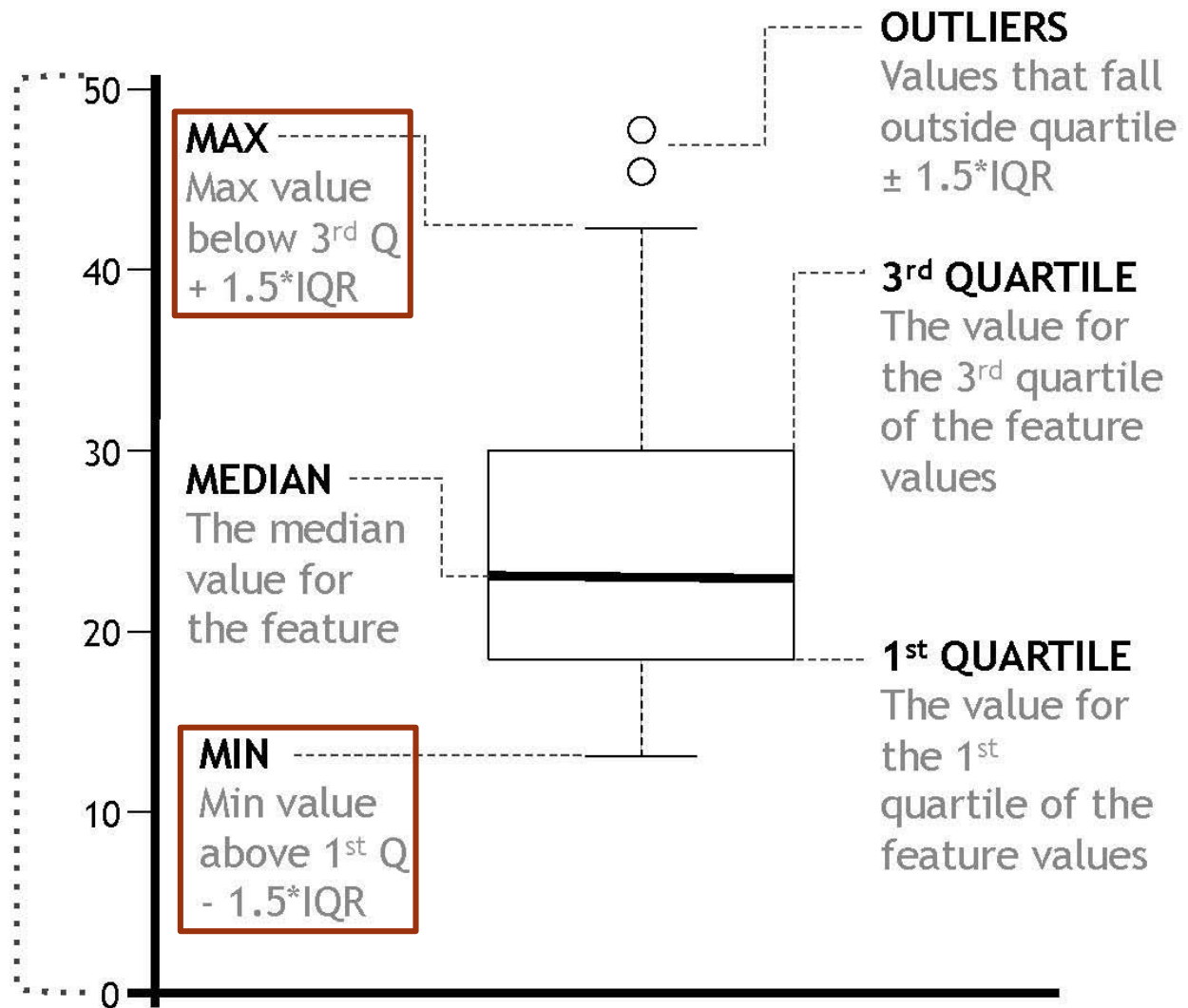


Figure A.8 (a) The structure of a box plot

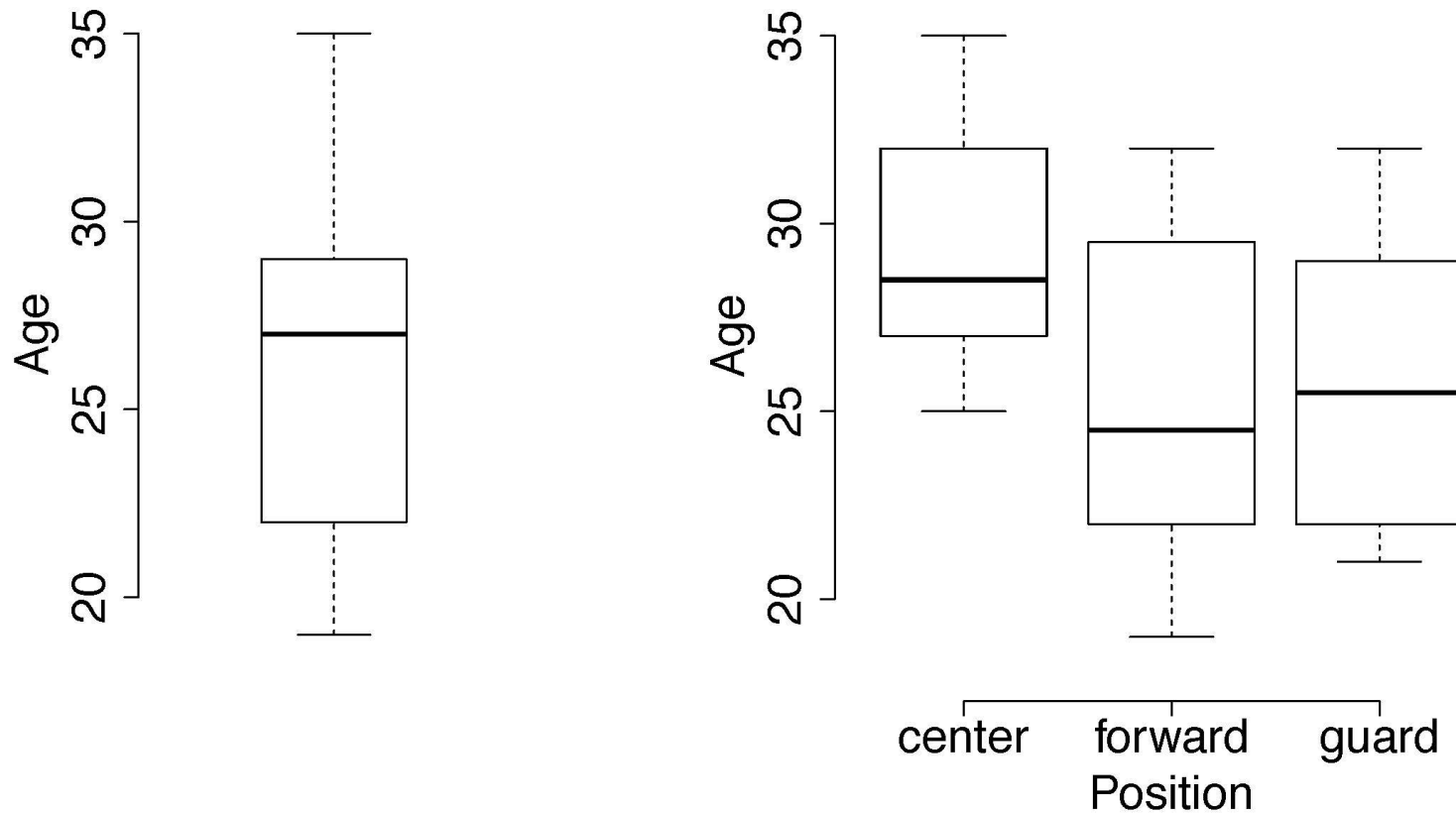


Figure 3.10-1: Using box plots to visualize the relationship between the **AGE** and the **POSITION** feature.

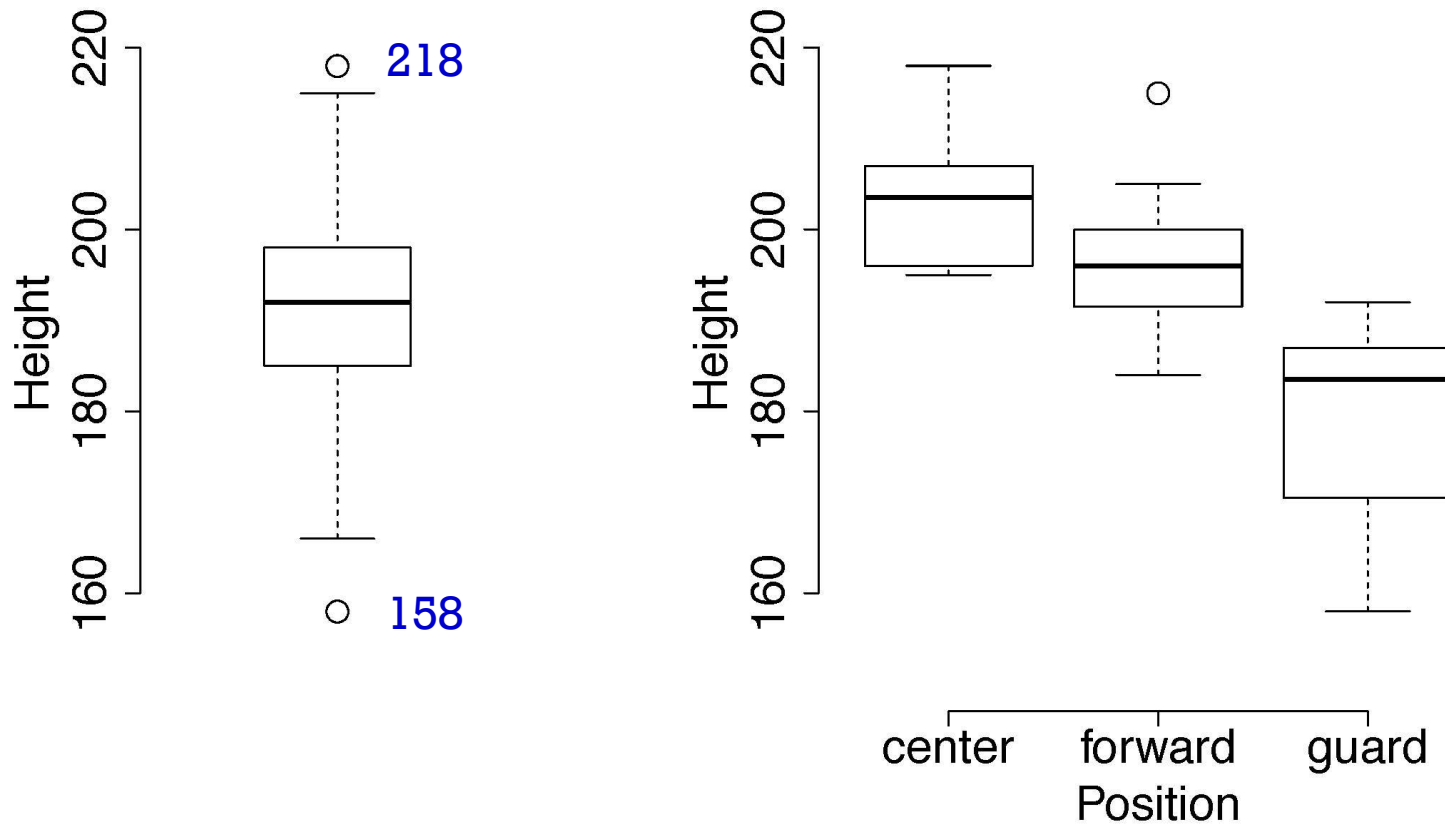


Figure 3.10-2: Using box plots to visualize the relationship between the **HEIGHT** feature and the **POSITION** feature.

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MEASURING COVARIANCE & CORRELATION

- As well as visually inspecting scatter plots, we can calculate formal measures of the relationship between **two continuous features** using **covariance** and **correlation**.

MEASURING COVARIANCE & CORRELATION

- For two features, a and b , in a dataset of n instances, the **sample covariance** between a and b is

$$\text{cov}(a, b) = \frac{1}{n-1} \sum_{i=1}^n \left((a_i - \bar{a}) \times (b_i - \bar{b}) \right) \quad (1)$$

where a_i and b_i are values of features a and b for the i th instance in a dataset, and $\bar{a}$ and $\bar{b}$ are the sample means of features a and b .

MEASURING COVARIANCE & CORRELATION

- **Covariance** values fall into the range $[-\infty, \infty]$:
 - **negative** values indicate a negative relationship
 - **positive** values indicate a positive relationship
 - values **near zero** indicate that there is little or no relationship between the features

$$\bar{a} = \frac{1}{n} \sum_{i=1}^n a_i$$

$$\text{var}(a) = \frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n-1}$$

$$\text{sd}(a) = \sqrt{\text{var}(a)} = \sqrt{\frac{\sum_{i=1}^n (a_i - \bar{a})^2}{n-1}}$$

- Calculating covariance between the **HEIGHT** feature, **WEIGHT** and **AGE** features from the basketball player

ID	HEIGHT (h)	$h - \bar{h}$	WEIGHT (w)	$w - \bar{w}$	$(h - \bar{h}) \times (w - \bar{w})$	AGE (a)	$a - \bar{a}$	$(h - \bar{h}) \times (a - \bar{a})$
1	192	0.9	218	3.0	2.7	29	2.6	2.3
2	218	26.9	251	36.0	967.5	35	8.6	231.3
3	197	5.9	221	6.0	35.2	22	-4.4	-26.0
4	192	0.9	219	4.0	3.6	22	-4.4	-4.0
5	198	6.9	223	8.0	55.0	29	2.6	17.9
			...					
26	191	-0.1	218	3.0	-0.3	19	-7.4	0.7
27	196	4.9	235	20.0	97.8	32	5.6	27.4
28	198	6.9	221	6.0	41.2	22	-4.4	-30.4
29	207	15.9	247	32.0	508.3	27	0.6	9.5
30	201	9.9	244	29.0	286.8	25	-1.4	-13.9
Mean	191.1		215.0			26.4		
Std Dev	13.6		19.8			4.2		
Sum					7,009.9			570.8

MEASURING COVARIANCE & CORRELATION

- Calculating covariance between the **HEIGHT** feature and the **WEIGHT** and **AGE** features from the basketball players dataset.

$$\text{cov}(\text{HEIGHT}, \text{WEIGHT}) = \frac{7,009.9}{29} = 241.72$$

$$\text{cov}(\text{HEIGHT}, \text{AGE}) = \frac{570.8}{29} = 19.7$$

MEASURING COVARIANCE & CORRELATION

- **Correlation** is a **normalized** form of covariance that ranges between -1 and +1.
- The correlation between two features, a and b , can be calculated as

$$\text{corr}(a, b) = \frac{\text{cov}(a, b)}{\text{sd}(a) \times \text{sd}(b)} \quad (2)$$

where $\text{cov}(a, b)$ is the covariance between features a and b and $\text{sd}(a)$ and $\text{sd}(b)$ are the standard deviations of a and b respectively.

MEASURING COVARIANCE & CORRELATION

- Correlation values fall into the range $[-1; 1]$:
 - values **close to -1** indicate a very strong negative correlation (or covariance)
 - values **close to 1** indicate a very strong positive correlation
 - values **around 0** indicate no correlation
- Features that have no correlation are said to be **independent**

MEASURING COVARIANCE & CORRELATION

- Calculating correlation between the **HEIGHT** feature and the **WEIGHT** and **AGE** features from the basketball players dataset.

$$\text{corr}(\text{HEIGHT}, \text{WEIGHT}) = \frac{241.72}{13.6 \times 19.8} = 0.898$$

$$\text{corr}(\text{HEIGHT}, \text{AGE}) = \frac{19.7}{13.6 \times 4.2} = 0.345$$

MEASURING COVARIANCE & CORRELATION

- In the majority of ABTs there are multiple **continuous** features between which we would like to explore relationships.
- Two tools that can be useful for this are the **covariance matrix** and the **correlation matrix**.

MEASURING COVARIANCE & CORRELATION

- The **covariance matrix**, usually denoted as Σ , between a set of continuous features, $\{a, b, \dots, z\}$, is given as

$$\Sigma_{\{a,b,\dots,z\}} = \begin{bmatrix} \text{var}(a) & \text{cov}(a,b) & \cdots & \text{cov}(a,z) \\ \text{cov}(b,a) & \text{var}(b) & \cdots & \text{cov}(b,z) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(z,a) & \text{cov}(z,b) & \cdots & \text{var}(z) \end{bmatrix} \quad (3)$$

MEASURING COVARIANCE & CORRELATION

- Similarly, the **correlation matrix** is just a normalized version of the covariance matrix and shows the correlation between each pair of features:

$$\text{correlation matrix}_{\{a,b,\dots,z\}} = \begin{bmatrix} \text{corr}(a,a) & \text{corr}(a,b) & \cdots & \text{corr}(a,z) \\ \text{corr}(b,a) & \text{corr}(b,b) & \cdots & \text{corr}(b,z) \\ \vdots & \vdots & \ddots & \vdots \\ \text{corr}(z,a) & \text{corr}(z,b) & \cdots & \text{corr}(z,z) \end{bmatrix} \quad (4)$$

MEASURING COVARIANCE & CORRELATION

- Calculating covariances matrix for the **HEIGHT** feature and the **WEIGHT** and **AGE** features from the basketball players dataset.

$$\Sigma_{\langle \text{Height}, \text{Weight}, \text{Age} \rangle} = \begin{bmatrix} 185.128 & 241.72 & 19.7 \\ 241.72 & 392.102 & 24.469 \\ 19.7 & 24.469 & 17.697 \end{bmatrix}$$

MEASURING COVARIANCE & CORRELATION

- Calculating correlation matrix for the **HEIGHT** feature and the **WEIGHT** and **AGE** features from the basketball players dataset.

$$\text{correlation matrix} = \begin{matrix} & \begin{matrix} \text{Height} \\ \text{Weight} \\ \text{Age} \end{matrix} \\ \begin{matrix} \text{Height} \\ \text{Weight} \\ \text{Age} \end{matrix} & \begin{bmatrix} 1.0 & 0.898 & 0.345 \\ 0.898 & 1.0 & 0.294 \\ 0.345 & 0.294 & 1.0 \end{bmatrix} \end{matrix}$$

MEASURING COVARIANCE & CORRELATION

- The **scatter plot matrix (SPLOM)** is really a visualization of the correlation matrix.
- This can be made more obvious by including the **correlation coefficients** in SPLOMs in the cells above the diagonal.

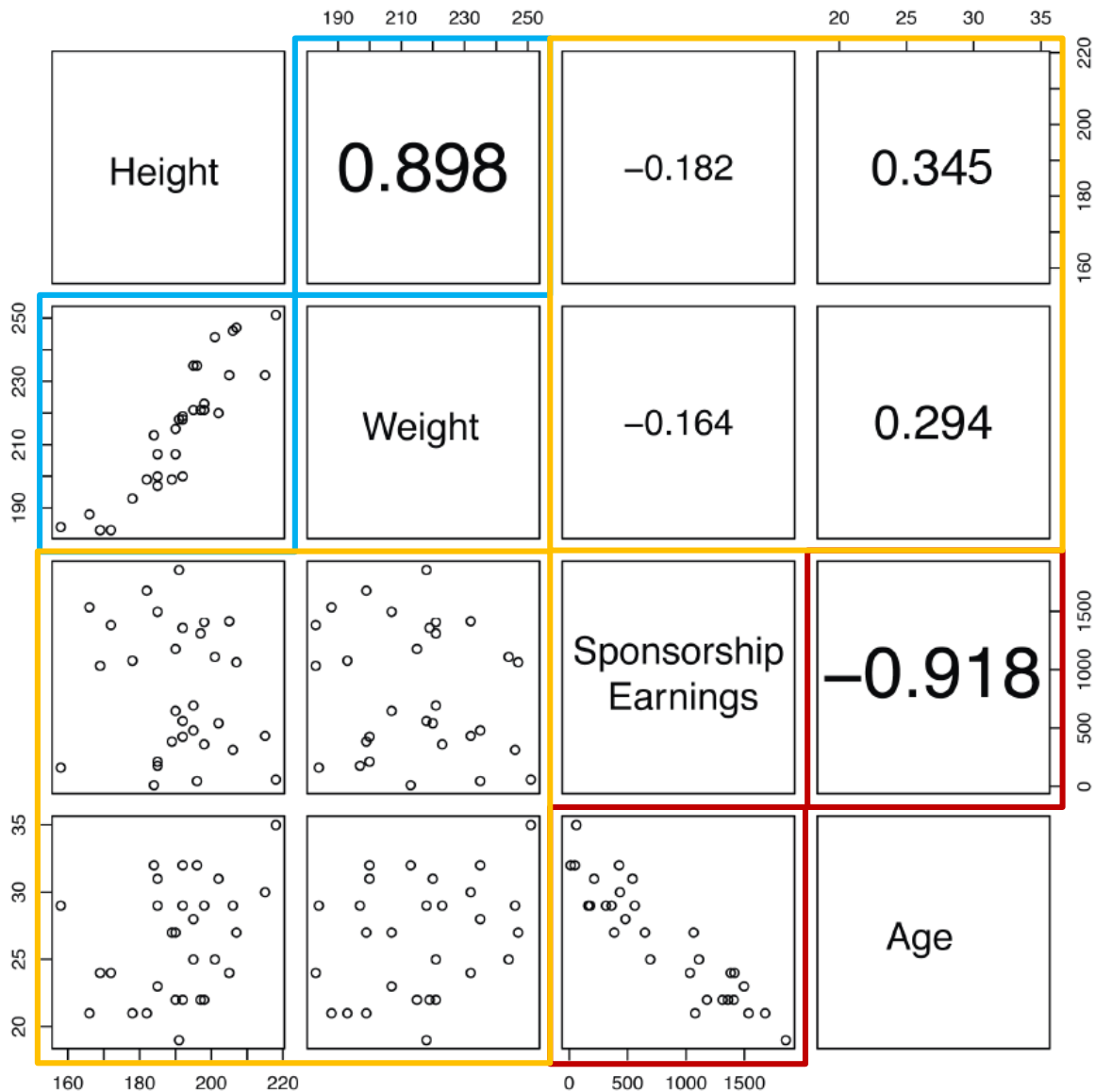


Figure 3.11 A scatter plot matrix showing scatter plots of the continuous features from the professional basketball team dataset in Table 3.7 with correlation coefficients included.

MEASURING COVARIANCE & CORRELATION

- Correlation is a good measure of the relationship between two continuous features, but it is **not by any means perfect**.
- Some of **the limitations** of measuring correlation are illustrated very clearly in the famous example of **Anscombe's quartet** by **Francis Anscombe**.

图表的重要性：ANSCOMBE的四组数据

- 1973年，统计学家F.J. Anscombe构造出了四组**奇特**的数据
- 它告诉人们，在分析数据之前，描绘数据所对应的图像有多么的重要！

Anscombe's Quartet

I		II		III		IV	
x	y	x	y	x	y	x	y
10.0	8.04	10.0	9.14	10.0	7.46	8.0	6.58
8.0	6.95	8.0	8.14	8.0	6.77	8.0	5.76
13.0	7.58	13.0	8.74	13.0	12.74	8.0	7.71
9.0	8.81	9.0	8.77	9.0	7.11	8.0	8.84
11.0	8.33	11.0	9.26	11.0	7.81	8.0	8.47
14.0	9.96	14.0	8.10	14.0	8.84	8.0	7.04
6.0	7.24	6.0	6.13	6.0	6.08	8.0	5.25
4.0	4.26	4.0	3.10	4.0	5.39	19.0	12.50
12.0	10.84	12.0	9.13	12.0	8.15	8.0	5.56
7.0	4.82	7.0	7.26	7.0	6.42	8.0	7.91
5.0	5.68	5.0	4.74	5.0	5.73	8.0	6.89

图表的重要性：ANSCOMBE的四组数据

- 这四组数据中， x 值的平均数都是9.0， y 值的平均数都是7.5
- x 值的方差都是10.0， y 值的方差都是3.75
- x 和 y 之间的相关度都是0.816，线性回归线都是 $y=3+0.5x$
- 单从这些统计数字上看来，四组数据所反映出的实际情况非常相近，而事实上，这四组数据有着天壤之别

图表的重要性：ANSCOMBE的四组数据

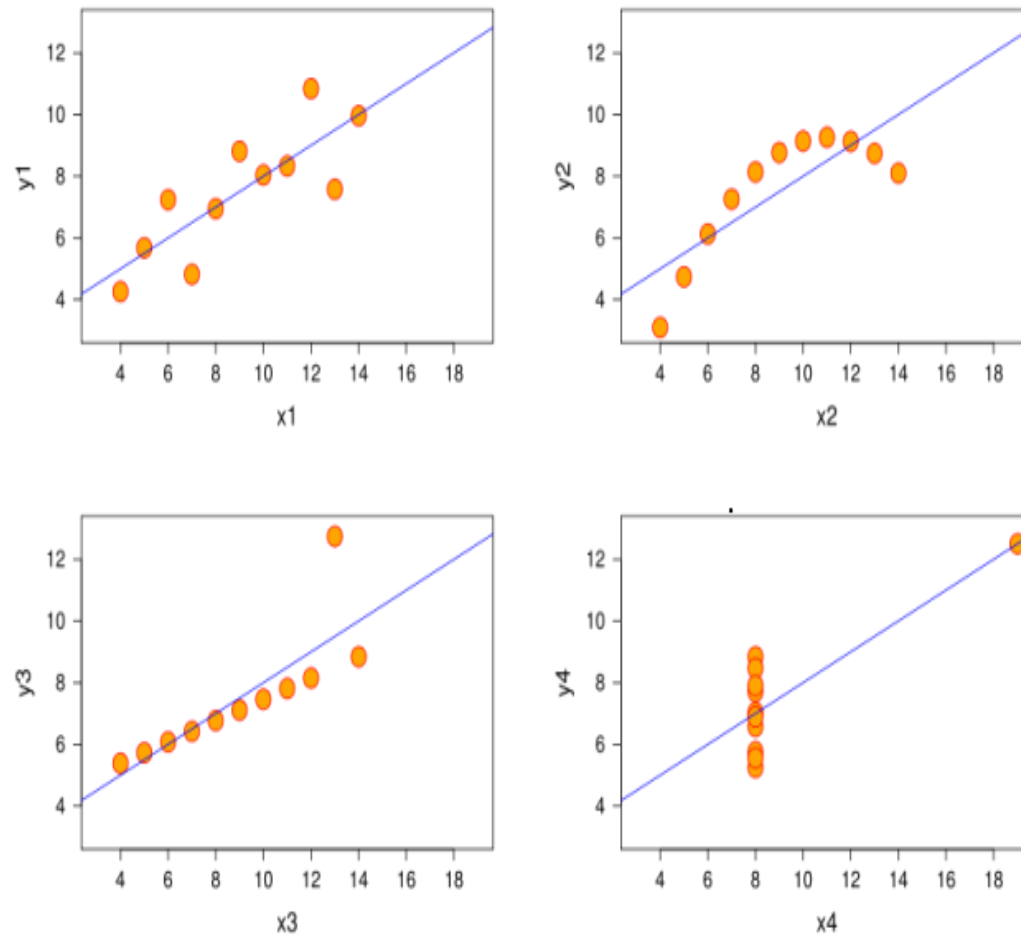


Figure 3.12 Anscombe's quartet. For all four samples, the correlation measure returns the same value (0.816) even though the relationship between the features is very different in each case.

MEASURING COVARIANCE & CORRELATION

- Perhaps the **most important** thing to remember in relation to correlation is that *correlation does not necessarily imply causation*.

MEASURING COVARIANCE & CORRELATION

- There are **two** main ways in which causation can be mistakenly assumed:
 - **Mistake the order** of a causal relationship
 - swallows cause hot weather
 - spinning windmills cause wind
 - playing basketball causes people to be tall
 - ...
 - ...

MEASURING COVARIANCE & CORRELATION

- There are **two** main ways in which causation can be mistakenly assumed:
 - ...
 - incorrectly infer causation between two features, **ignore a third important, but hidden, feature**
 - A famous example: In 1999, an article was published in “**Nature**” outlining a causal relationship between young children sleeping with a night-light turned on and these children developing short-sightedness in later life.
 - Later studies could not replicate this link...
 - This is an example of a **confounding feature**, a feature that influences two others and so leads to the appearance of a causal relationship.

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- Some data preparation techniques change the way data is represented just to make it **more compatible** with certain machine learning algorithms.
 - **Normalization**
 - **Binning**
 - **Sampling**

1. NORMALIZATION

- **Normalization** techniques can be used to change a **continuous feature** to fall within a specified range while maintaining the relative differences between the values for the feature.

1. NORMALIZATION

- We use **range normalization** to convert a feature value into the range [low; high] as follows:

$$a'_i = \frac{a_i - \min(a)}{\max(a) - \min(a)} \times (high - low) + low \quad (5)$$

1. NORMALIZATION

- Another way to normalize data is to **standardize** it into **standard scores**.
- A standard score measures how many standard deviations a feature value is from the mean for that feature.
- We calculate a standard score as follows:

$$a'_i = \frac{a_i - \bar{a}}{sd(a)} \quad (6)$$

- The result of normalizing a **small sample** of the **HEIGHT** and **SPONSORSHIP EARNINGS** features from the professional basketball squad dataset.

	HEIGHT			SPONSORSHIP EARNINGS		
	Values	Range	Standard	Values	Range	Standard
	192	0.500	-0.073	561	0.315	-0.649
	197	0.679	0.533	1,312	0.776	0.762
	192	0.500	-0.073	1,359	0.804	0.850
	182	0.143	-1.283	1,678	1.000	1.449
	206	1.000	1.622	314	0.164	-1.114
	192	0.500	-0.073	427	0.233	-0.901
	190	0.429	-0.315	1,179	0.694	0.512
	178	0.000	-1.767	1,078	0.632	0.322
	196	0.643	0.412	47	0.000	-1.615
	201	0.821	1.017	1111	0.652	0.384
Max	206			1,678		
Min	178			47		
Mean	193			907		
Std Dev	8.26			532.18		

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2. BINNING

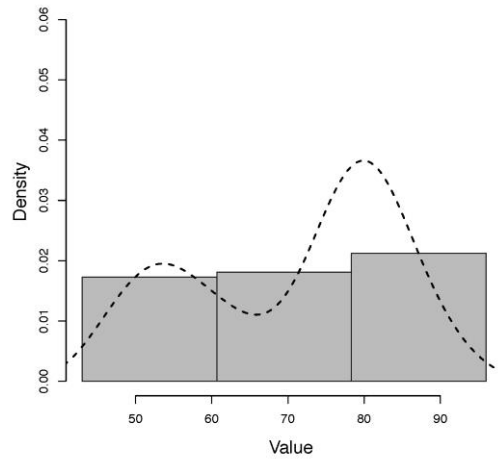
- **Binning** involves converting a **continuous** feature into a **categorical** feature.
- To perform binning, we define a series of ranges (called **bins**) for the continuous feature that correspond to the levels of the new categorical feature we are creating.
- We will introduce two of the more popular ways of defining bins:
 - **equal-width binning**
 - **equal-frequency binning**

Breaking Out the Move-In Bins at Barnard

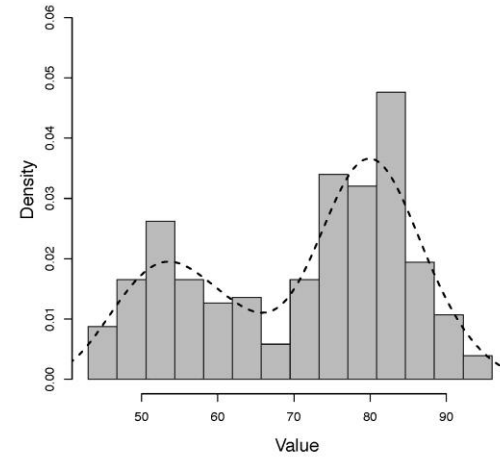


2. BINNING

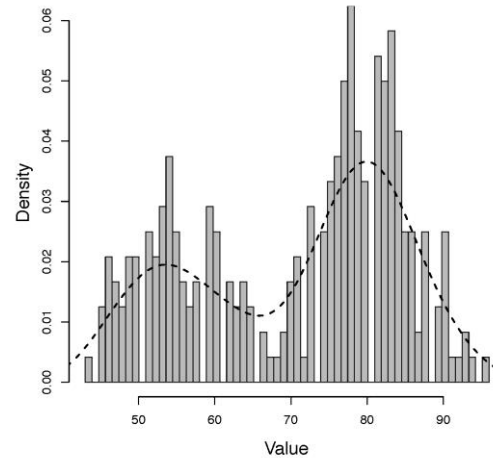
- Deciding on the number of bins can be difficult.
- The general **trade-off** is this:
 - If we set the number of bins to a **very low** number we may lose a lot of information
 - If we set the number of bins to a **very high** number then we might have very few instances in each bin or even end up with empty bins.



(a) 3 bins



(b) 14 bins

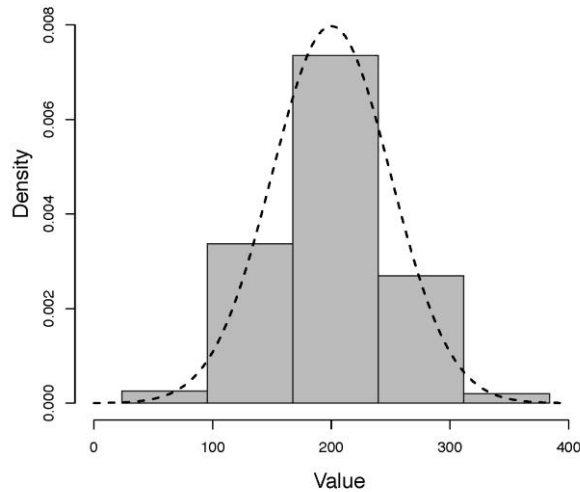


(c) 60 bins

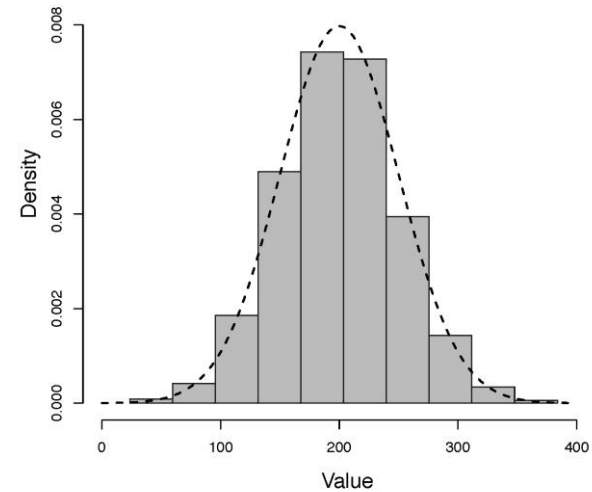
Figure 3.13 The effect of using different numbers of bins when using binning to convert a continuous feature into a categorical feature.

2. BINNING

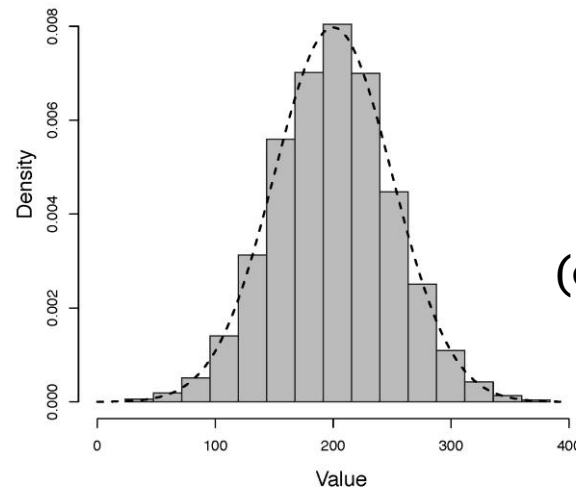
- The **equal-width binning** algorithm splits the range of the feature values into b bins each of size $\frac{range}{b}$.



(a) 5 Equal-width bins



(b) 10 Equal-width bins

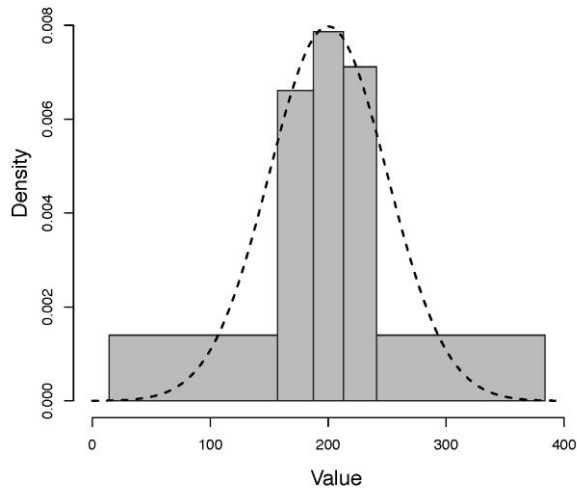


(c) 15 Equal-width bins

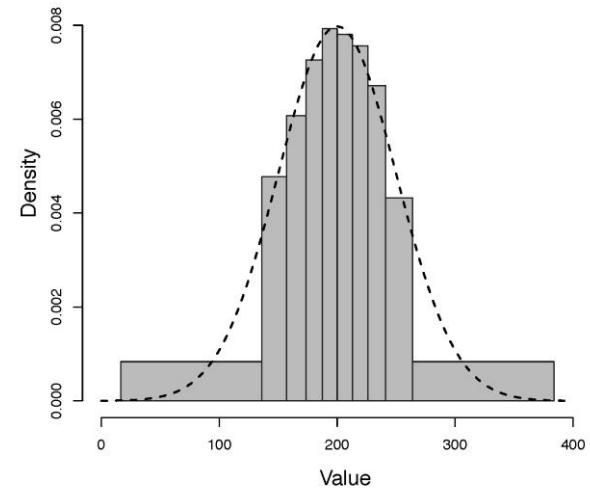
Figure 3.14-1 (a)–(c) **Equal-width binning** of normally distributed data with different numbers of bins. The dashed lines illustrate the distribution of the original continuous feature values, and the gray boxes represent the bins.

2. BINNING

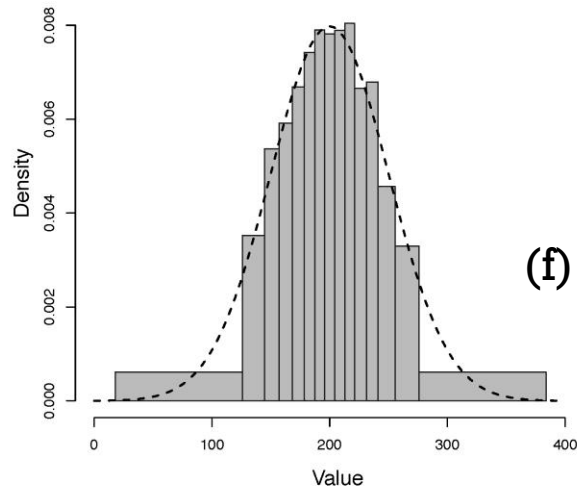
- **Equal-frequency binning** first sorts the continuous feature values into ascending order and then places an equal number of instances into each bin, starting with bin 1.
- The number of instances placed in each bin is simply the total number of instances divided by the number of bins, b .



(d) 5 Equal-frequency bins



(e) 10 Equal-frequency bins



(f) 15 Equal-frequency bins

Figure 3.14-2 (d)–(f) the same data binned into the same number of bins using [equal-frequency binning](#). The dashed lines illustrate the distribution of the original continuous feature values, and the gray boxes represent the bins.

CONTENTS

- **Advanced Data Exploration**

- Visualizing Relationships Between Features
- Measuring Covariance & Correlation

- **Data Preparation**

- Normalization
- Binning
- Sampling

- **Summary**

3. SAMPLING

- Sometimes the dataset is **so large** that we do not use all the data available in an ABT and instead **sample** a smaller percentage from the larger dataset.
- We need to be careful when sampling, however, to ensure that the resulting datasets are still **representative** of the original data and that no unintended **bias** is introduced during this process.

3. SAMPLING

- Common forms of sampling include:
 - **top sampling**
 - **random sampling**
 - **stratified sampling**
 - **under-sampling**
 - **over-sampling**

3. SAMPLING

Top sampling

- **Top sampling** simply selects the top $s\%$ of instances from a dataset to create a sample.
- Top sampling runs a **serious risk** of introducing bias, however, as the sample will be affected by any ordering of the original dataset.
- We recommend that top sampling be **avoided**.

3. SAMPLING

Random sampling

- Our recommended default, **random sampling** randomly selects a proportion of $s\%$ of the instances from a large dataset to create a smaller set.
- Random sampling is a **good choice** in most cases as the random nature of the selection of instances should avoid introducing bias.

3. SAMPLING

Stratified sampling

- **Stratified sampling** is a sampling method that ensures that the **relative frequencies** of the levels of a specific **stratification feature** are maintained in the sampled dataset.
- To perform stratified sampling:
 - the instances in a dataset are divided into groups (or strata), where each group contains only instances that have a particular level for the stratification feature
 - $s\%$ of the instances in each stratum are randomly selected
 - these selections are combined to give an overall sample of $s\%$ of the original dataset.

3. SAMPLING

Under-sampling/Over-sampling

- In contrast to stratified sampling, sometimes we would like a sample to contain **different relative frequencies** of the levels of a particular feature to the distribution in the original dataset.
- To do this, we can use **under-sampling** or **over-sampling**.

3. SAMPLING

Under-sampling

- **Under-sampling** begins by dividing a dataset into groups, where each group contains only instances that have a particular level for the feature to be under-sampled.
- The number of instances in **the smallest group** is the under-sampling **target size**.
- Each group containing more instances than the smallest one is then randomly sampled by the appropriate percentage to create a subset that is the under-sampling target size.
- These under-sampled groups are then combined to create the overall under-sampled dataset.

3. SAMPLING

Over-sampling

- **Over-sampling** addresses the same issue as under-sampling but in the opposite way around.
- After dividing the dataset into groups, the number of instances in **the largest group** becomes the over-sampling **target size**.
- From each smaller group, we then create a sample containing that number of instances using **random sampling with replacement**.
- These larger samples are combined to form the overall over-sampled dataset.

CONTENTS

- **Advanced Data Exploration**

- Visualizing Relationships Between Features
- Measuring Covariance & Correlation

- **Data Preparation**

- Normalization
- Binning
- Sampling

- **Summary**

- The key outcomes of the **data exploration** process are that the practitioner should
 - Have gotten to know the features within the ABT (Analytics Base Table) , especially their **central tendencies, variations, and distributions**.
 - Have identified any **data quality issues** within the ABT, in particular **missing values, irregular cardinality, and outliers**.
 - Have corrected any data quality issues due to **invalid data**.
 - Have recorded any data quality issues due to **valid data** in a **data quality plan** along with potential handling strategies.
 - Be confident that enough good quality data exists to continue with a project.

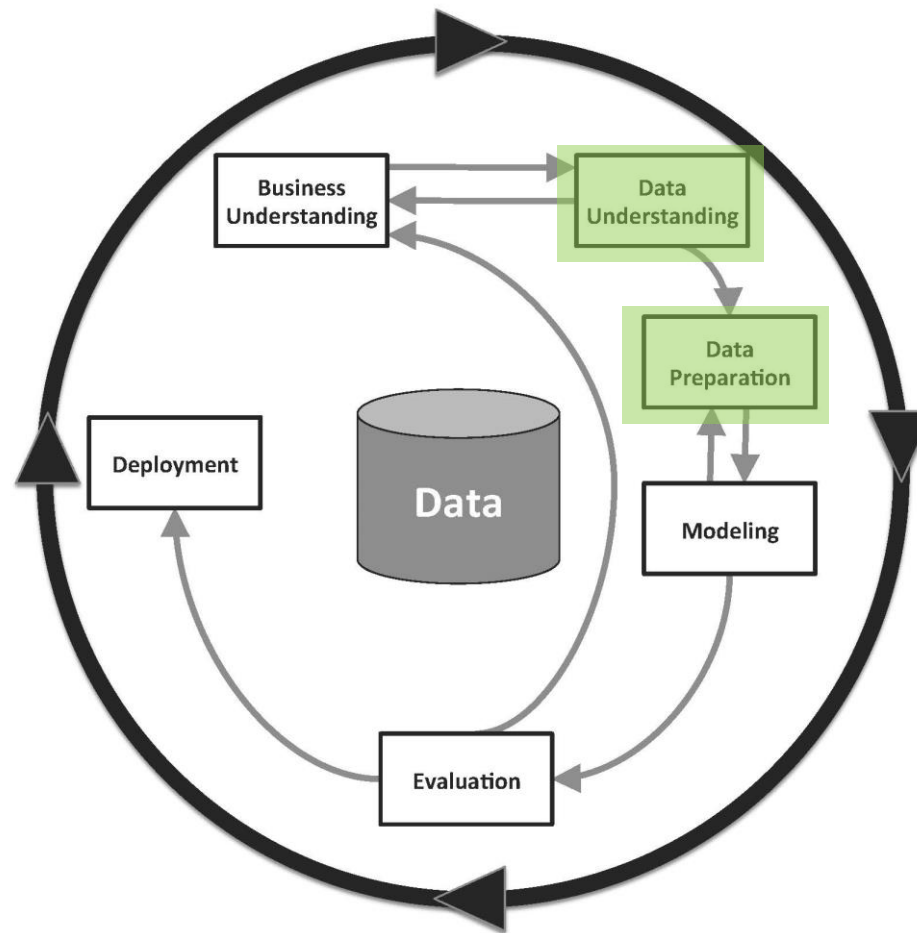
■ Advanced Data Exploration

- Visualizing Relationships Between Features
 - Scatter plot- continuous- continuous
 - Scatter plot matrix (SPLOM)
 - Small multiple bar plot categorical- categorical
 - Stacked bar plot
 - Small multiple histogram categorical- continuous
 - Box plot
- Measuring Covariance & Correlation

■ Data Preparation

- Normalization
 - Range normalization
 - Standard scores
- Binning
 - equal-width binning
 - equal-frequency binning
- Sampling
 - top sampling
 - random sampling
 - stratified sampling
 - under-sampling
 - over-sampling

- **Data exploration** is a key part of both the **Data Understanding** and, **Data Preparation** phases of CRISP-DM.



HOMEWORK

- Reading: Section 3.5~3.7
- Exercises: 3.9-1, 2, 5, 6

成绩评定

- 平时成绩50%
 - 习题 20%
 - 第一次作业8% (第1章-第4章)
 - 第二次作业12% (第5章-第9章)
 - 论文+口头报告 30%
 - 报告15%
 - 论文15%
- 期末考核成绩50%
 - 2027年1月6日闭卷考试 (英文试卷)